Lab 3

Parameter Identification

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Group 06

Team members:
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Team Contribution Table:

Team member	Experimental work and data collection	Analysis	MATLAB/ Simulink Code	Interpretation and Validation	Report Writing and Editing	Overall Contribution
Member 1 (Saud Amjad):	**S.A.**	**S.A.**	**S.A.**	**S.A.**	**S.A.**	33.33%
Member 2 (Hamzah Nadeem):	**H.N**	**H.N**	**H.N**	**H.N**	**H.N**	33.34%
Member 3 (Temuulen Ankhbayar):	**T.A**	**T.A**	**T.A**	**T.A**	**T.A**	33.33%
TOTAL:						100%

Acknowledgement: By submitting this report, all team members confirm that the information above accurately represents their contributions and that all team members have reviewed and approved the final report and associated files

Table 1: Results Summary

Quantity	Your Value	Where the Evidence is
Positive Coulomb Friction Voltage Vc+ (V)	0.19	6.1.3 (Page 3)
Negative Coulomb Friction Voltage Vc- (V)	-0.19	6.1.4 (Page 4)
Total Viscous Coefficient from the Fit (V·s/rad)	-2.3×10^{-3}	6.1.8 (Page 6)
Viscous friction coefficient μf (V·s/rad)	-0.001261	6.3.7 (Page 13)
Average Motor Resistance Rm (Ω)	9.8862	6.3.6 (Page 12)
Average Back-EMF Constant km (V·s/rad)	0.040961	6.3.7 (Page 12)
Experimental Transfer Function Ω(s)/Vm(s)	$\frac{1}{[0.0049192225s + 0.040961]}$	6.3.8 (Page 15)

6.1: Friction Identification:

3. Reported V_{C+} Voltage: **0.19V.** The motor would reliably start rotating at 0.2V when manually incrementing the input voltage by 0.01V. Although the plot with the ramp function shows movement (non-zero angular velocity) before 0.2V, we observed it was only after V_C exceeded 0.19V, the motor would reliably spin, which means it guarantees overcoming the static friction of the motor.

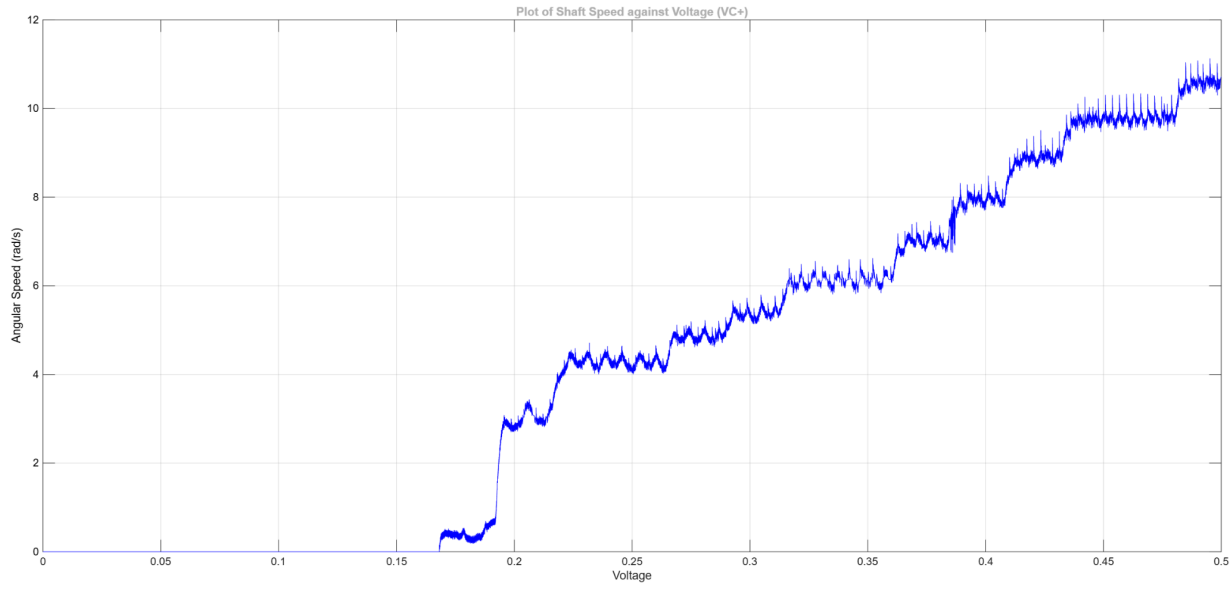


Figure 01: Plot of Shaft Speed against Voltage (V_{C+}) - Implemented with Ramp Function

4. Reported V_{C-} Voltage: **-0.19 V.** The motor would reliably start rotating at -0.2V, but at some voltages below that (ex. -0.16 V), the motor would partially spin and then stop, as seen in the graph below. It was decided to use -0.19 V as V_{C-} because the motor would reliably spin at -0.20V, which means it guarantees overcoming the static friction of the motor.

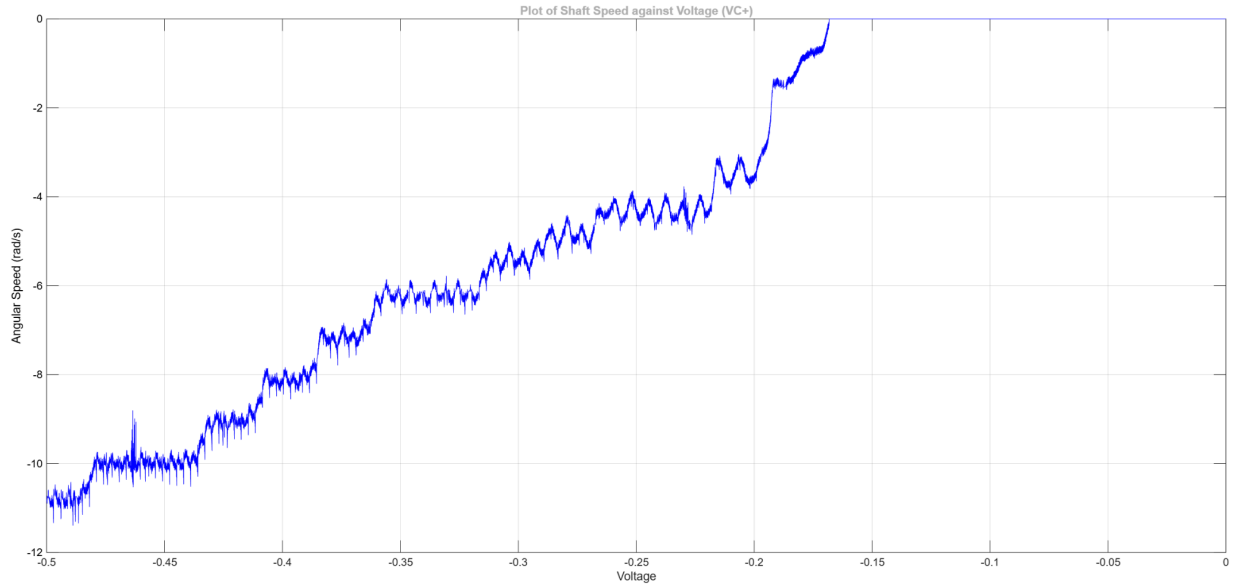


Figure 02: Plot of Shaft Speed against Voltage (V_{C-}) - Implemented With Ramp Function

5.

Table 2: Measuring Speed from Specific Voltages

Applied Voltage: V_m (V)	Measured Speed: ω_m (rad/s)
-4	-97
-3	-72.25
-2	-46.8
-1	-23.9
-0.19	0
0.19	0
1	23.4
2	46.5
3	71.5
4	96.5

6.

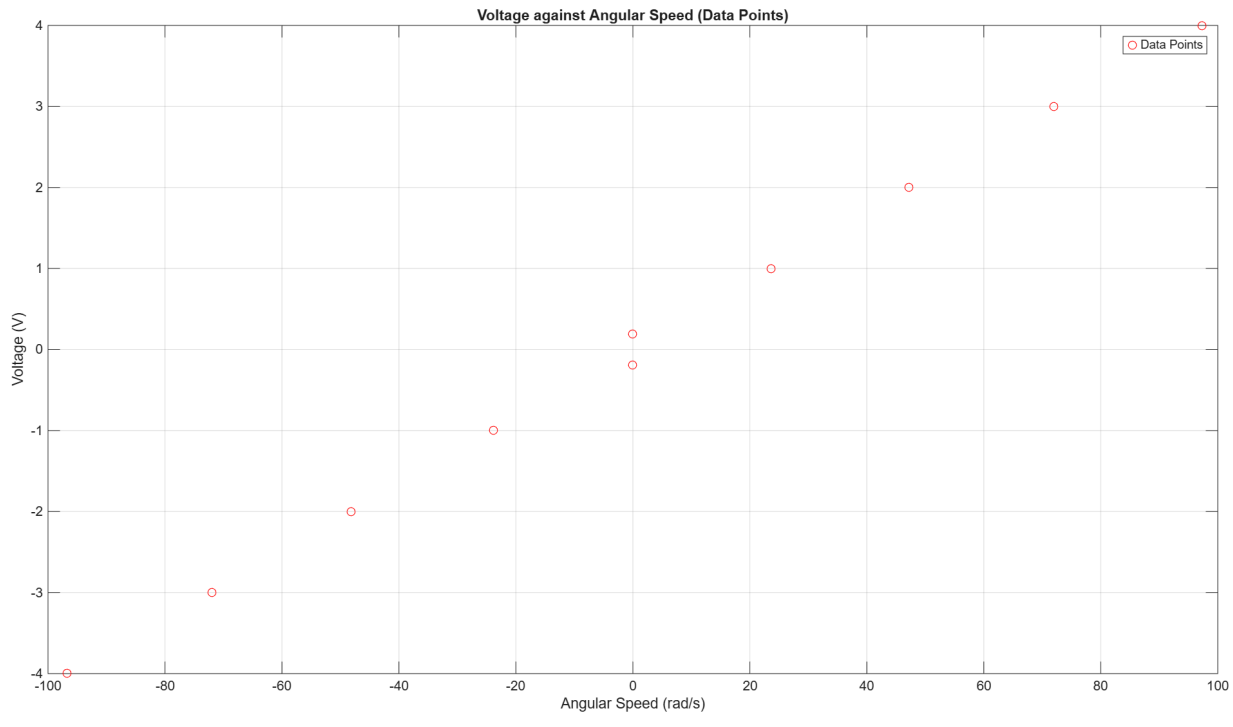


Figure 03: Plot of Voltage against Measured Velocity

7.

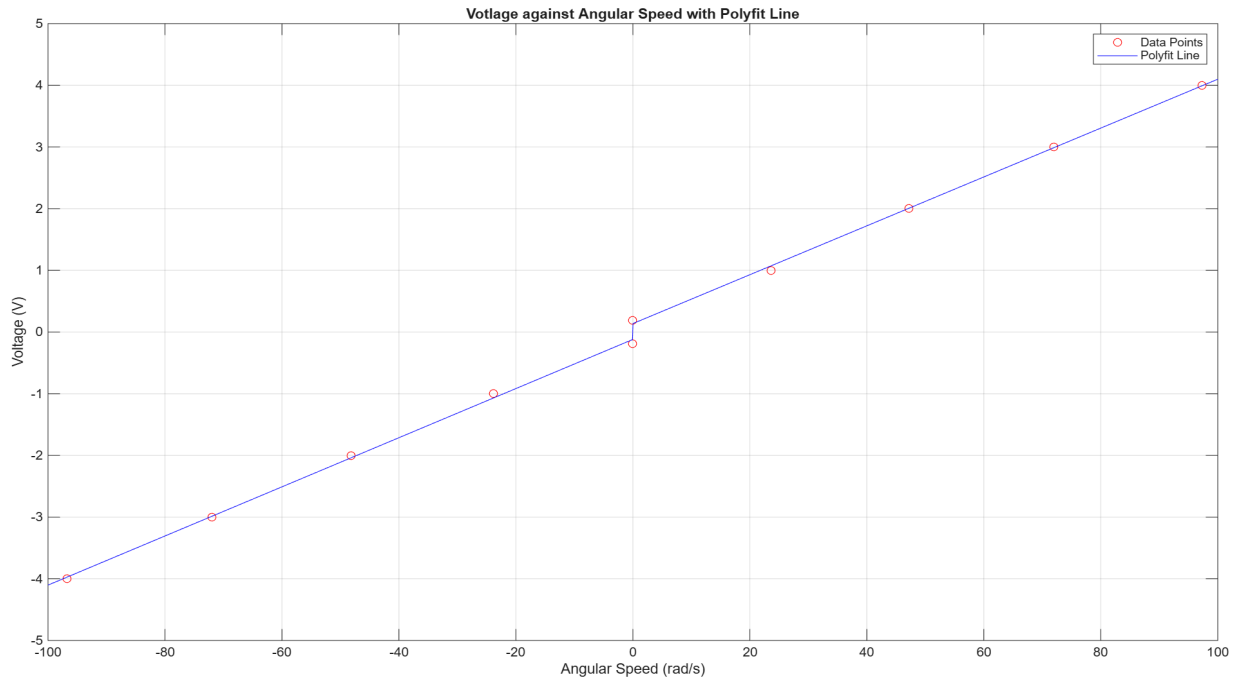


Figure 04: Plot of Voltage against Measured Velocity (Estimated VS Measured)

```
>> Lab03_Group06_livescript

coeffs_n =

    0.0398   -0.1197


coeffs_p =

    0.0396    0.1371
```

$x = Angular\ Velocity\ [\omega]\ (rad/s)$
$y = Applied\ Input\ Voltage\ [V_m]\ (V)$

Using MATLAB Polyfit function, the following polynomial expressions were derived:

$For\ x < 0$:

$$y = 0.0398x - 0.1197$$

$For\ x > 0$:

$$y = 0.0396x + 0.1371$$

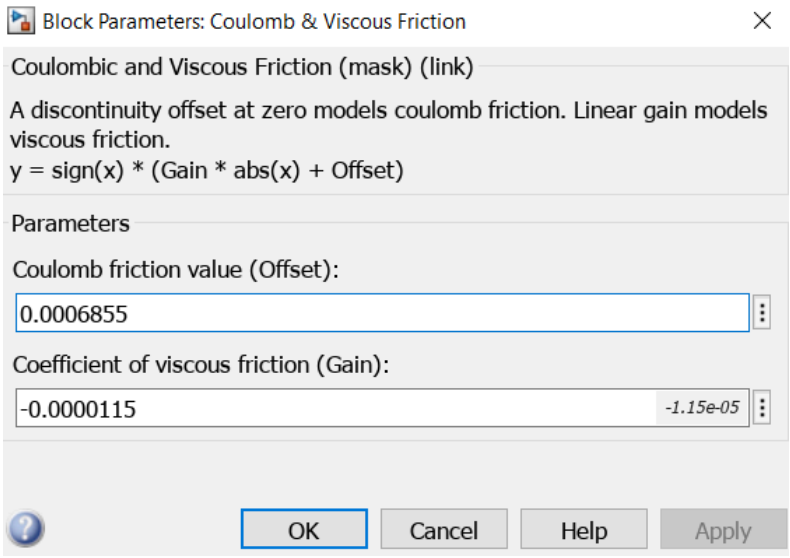
8. The value of the coefficient we identified is: $-\ 2.3 \times 10^{-3}\ V/(rad/s)$. Since the slope value is a combination of the viscous friction coefficient and the back-emf constant, we subtract the specification sheet back-emf constant (k_m) from the slope to get our viscous friction coefficient (μ_f = slope - k_m). We took the average of the two slopes found above and used the result to determine μ_f:

$$\overline{slope} = \frac{slope_{(x>0)} + slope_{(x<0)}}{2} = \frac{(0.0398) + (0.0396)}{2} = 0.0397\ V/(rad/s)$$

$$\mu_f = (slope - k_{m(datasheet)}) = (0.0397) - (0.042) = -\ 2.3 \times 10^{-3}\ V/(rad/s)$$

6.2: Model Friction Validation:

2. The value we selected for our Coulomb and Viscous Friction block were:



Below are the calculations used to determine the input parameters above:

From the lab manual, provided that we know the motor angular velocity, ω_m, we can use the following equation to model friction:

$$F_f = \mu\omega_m + F_c\, sgn(\omega_m)$$

Which can be rewritten as:

$$F_f = sgn(\omega_m) \cdot (\mu \cdot |\omega_m| + F_c)$$

Note that this equation is also represented in the above Coulomb Friction Block, where the Coulomb friction value (Offset) is F_C and the viscous friction coefficient (Gain) is μ. The lab manual also references two more equations to determine F_c and μ from the determined values for V_c and μ_f.

$$V_c = \frac{R_m}{k_t} F_c \rightarrow F_c = \frac{k_t}{R_m} V_c$$

$$\mu_f = \frac{R_m}{k_t}\mu \rightarrow \mu = \frac{k_t}{R_m}\mu_f$$

Coulomb Friction Value (Offset):

$$F_c = \frac{k_t}{R_m} V_c, \text{ where } V_c = max(V_{c+}, V_{c-})$$

$$F_c = \frac{0.042\ N{\cdot}m/A}{8.4\ \Omega}(0.1371\ V)$$

$$F_c = 6.855 \times 10^{-4}\ N \cdot m$$

Coefficient of Viscous Friction (Gain):

$$\mu_f = \frac{R_m}{k_t}\mu$$

$$\mu = \frac{k_t}{R_m}(slope - k_m)$$

$$\mu = \frac{0.042\ N{\cdot}m/A}{8.4\ \Omega}(0.0397\ V/(rad/s) - 0.042\ V/(rad/s))$$

$$\mu = -1.15 \times 10^{-5}\ \frac{N{\cdot}m{\cdot}s}{rad}$$

5.

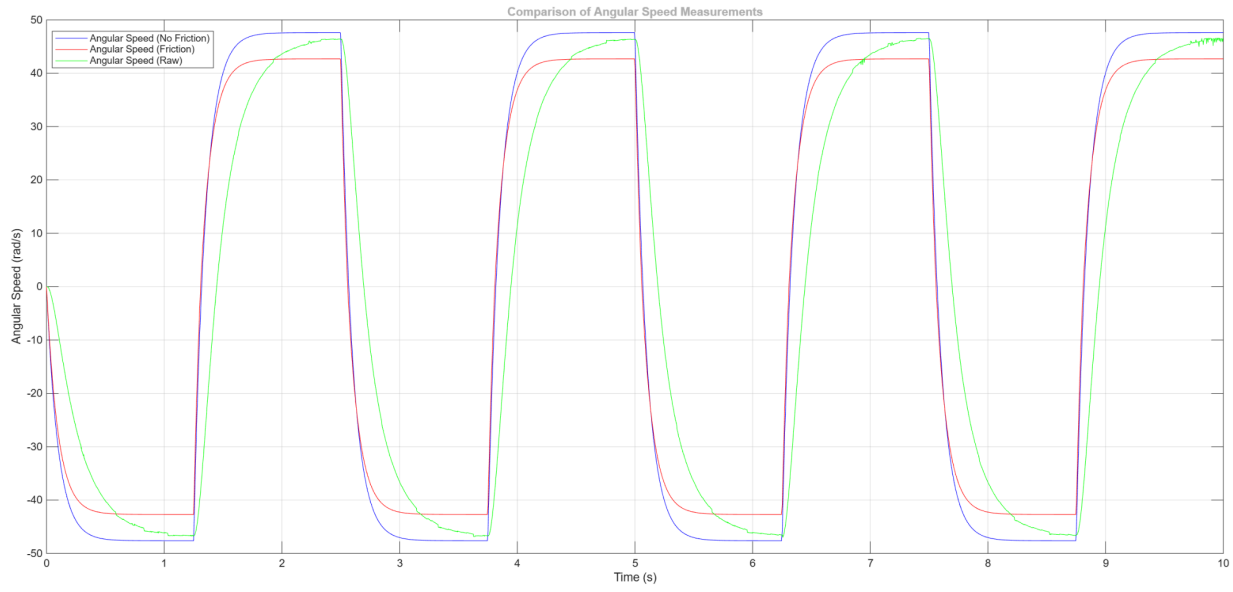


Figure 05: Plot of Angular Speed With Friction, Without Friction, and Physical Motor Speed

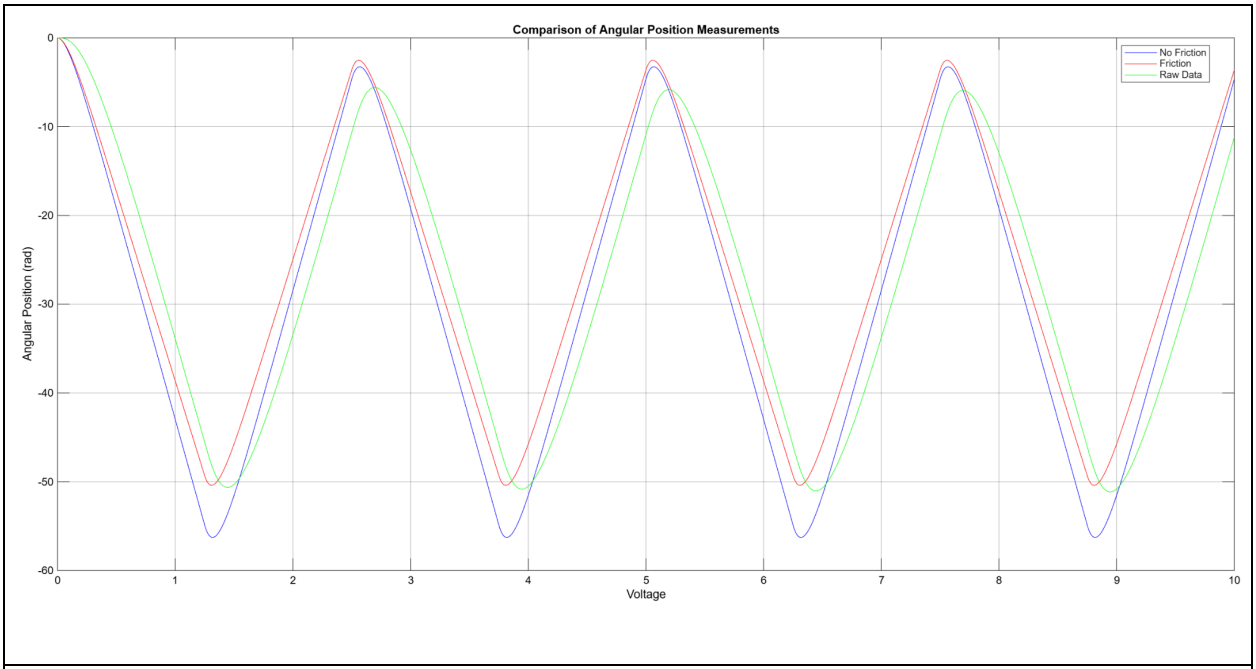


Figure 06: Plot of Position With Friction, Without Friction, and Physical Motor Position

Note that for Figure 06, we did not use the integrator block in Simulink. The motor position graph was obtained by importing the angular speed plot data into MATLAB, and then using the MATLAB *cumsum* function to integrate over the data. The code for this is added to the attached MATLAB Livescript File (.mlx).

6.

Including friction did not improve the accuracy of our system. This is because our μ_f ended up being a negative value after calculating the difference between the slope of the fitted line and the k_m value. For the k_m value, we used the specification sheet value of 0.042 (V/(rad/s)). If we used a k_m value more representative of our provided motor in the lab, then the friction model could be more accurate than the no friction model.

6.3: Motor Resistance and back-emf Constant Identification:

1. Deriving R_m:

From the experiment, by stalling the motor, the speed $\omega_m(t)$ is 0 rad/s, so the term $k_m \omega_m(t)$ is equal to 0. From there, we can isolate for R_m and solve it with the recorded values:

$$v_m(t) - R_m i_m(t) - k_m \omega_m(t) = 0,\ \omega_m(t) = 0$$

$$v_m(t) - R_m i_m(t) = 0$$

$$v_m(t) = R_m i_m(t)$$

$$\boxed{R_m = \frac{v_m(t)}{i_m(t)}}$$

Assumptions: One assumption was as the inductance of the motor, L_m, was much less than the resistance of the motor, R_m, it can be ignored. Another assumption was as the motor was being stalled (stopped by holding the motor shaft stationary by grasping the inertial disc), the angular speed (ω) of the motor was exactly 0 rad/s. The angular speed is more likely to be extremely close to 0 but not exact, due to our hand not being able to exactly counter the force produced from the motor.

2. Deriving k_m:

$$v_m(t) - R_m i_m(t) - k_m \omega_m(t) = 0$$

$$v_m(t) - R_m i_m(t) = k_m \omega_m(t)$$

$$\boxed{k_{m(experimental)} = \frac{v_m(t) - R_m i_m(t)}{\omega_m(t)}}$$

Assumptions: Again, one assumption was as the inductance of the motor, L_m, was much less than the resistance of the motor, R_m, so it can be ignored. Note in the above equation that $k_{m(experimental)}$ value is a summation of the back-emf constant k_m and the viscous friction coefficient μ_f. Because they both effectively behave as viscous friction, there is no way to determine the individual values. Thus, we will treat this value as k_m, and in a later step (6.3.7), we will use this value to solve for $\mu_{f(experimental)}$

4.

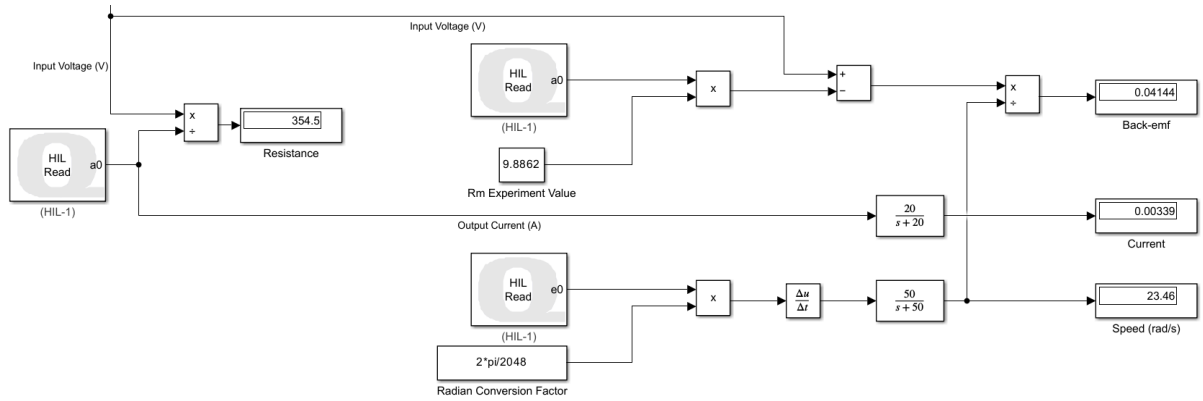


Figure 07: Block Diagram to find R_m and k_m

At a user-defined input voltage value, we are using the HIL read blocks to read the current and angular speed, which can be used with the input voltage value to calculate for the resistance of the motor (R_m) and the back-emf constant (k_m).

5.

Table 3: Results from R_m experiment

Applied Input Voltage: V_m (V)	Measured Current: I_m (A)	Resistance: R_m (Ω)
-5	-0.5355	9.337
-4	-0.4195	9.536
-3	-0.3046	9.848
-2	-0.1918	10.430
-1	-0.09872	10.130
1	0.09751	10.260
2	0.189	10.580
3	0.3038	9.875
4	0.4195	9.536
5	0.5359	9.330

6.

Table 4: Comparison of Results from R_m experiment

Average R_m value (Ω):	9.8862
Data sheet R_m value (Ω):	8.4

There is a moderate discrepancy between the experimental value for the average, R_m, to the datasheet value of R_m. The experimental value was 17.69% larger than the datasheet value.

7. Table 5: Results from k_m experiment

Applied Voltage: V_m (V)	Measured speed: ω_m (rad/s)	Measured Current: I_m (A)	Back-emf: k_m (V· s/rad)
-5	-121.5	-0.003223	0.0409
-4	-97.64	-0.003223	0.04064
-3	-72.45	-0.003626	0.04091
-2	-46.95	-0.003703	0.04184
-1	-23.76	-0.004354	0.04008
1	23.36	0.003652	0.04144
2	46.88	0.003257	0.04207
3	72.02	0.003542	0.04088
4	97.23	0.004446	0.04073
5	122.7	0.005239	0.04012

Table 6: Comparison of Results from k_m experiment

Average k_m value (V· s/rad):	0.040961
Spec sheet k_m value (V· s/rad):	0.042

The two values are reasonably close, with the discrepancy between the experimental value for the average, k_m, to the datasheet value of k_m was calculated to be small. The experimental value was only 2.47% smaller than the datasheet value. This discrepancy could be due to the fact that each motor has some variations and that the value on the datasheet is an average value over multiple motors.

Now that the experimental k_m has been determined, we can use this value and calculate our experimental viscous friction coefficient, $\mu_{f\ (experimental)}$:

$$\mu_{f\ (experimental)} = \overline{slope}_{(from\ 6.2)} - \overline{k}_{m(experimental)}$$

$$\mu_{f\ (experimental)} = (0.0397\ V/(rad/s)) - (0.040961\ V/(rad/s))$$

$$\boxed{\mu_{f\ (experimental)} = -0.001261\ V/(rad/s)}$$

Where $\overline{k}_{m(experimental)}$ represents the average of the 10 $k_{m(experimental)}$ values determined.

There are a few reasons for why this value is negative, such as a degree of uncertainty in the slope fitted to our measured values in Section 6.2 but also from the same assumption made in 6.3.2 for $k_{m(experimental)}$. In reality, $k_{m(experimental)}$ was a combined value of the summation of k_m and μ_f.

8.

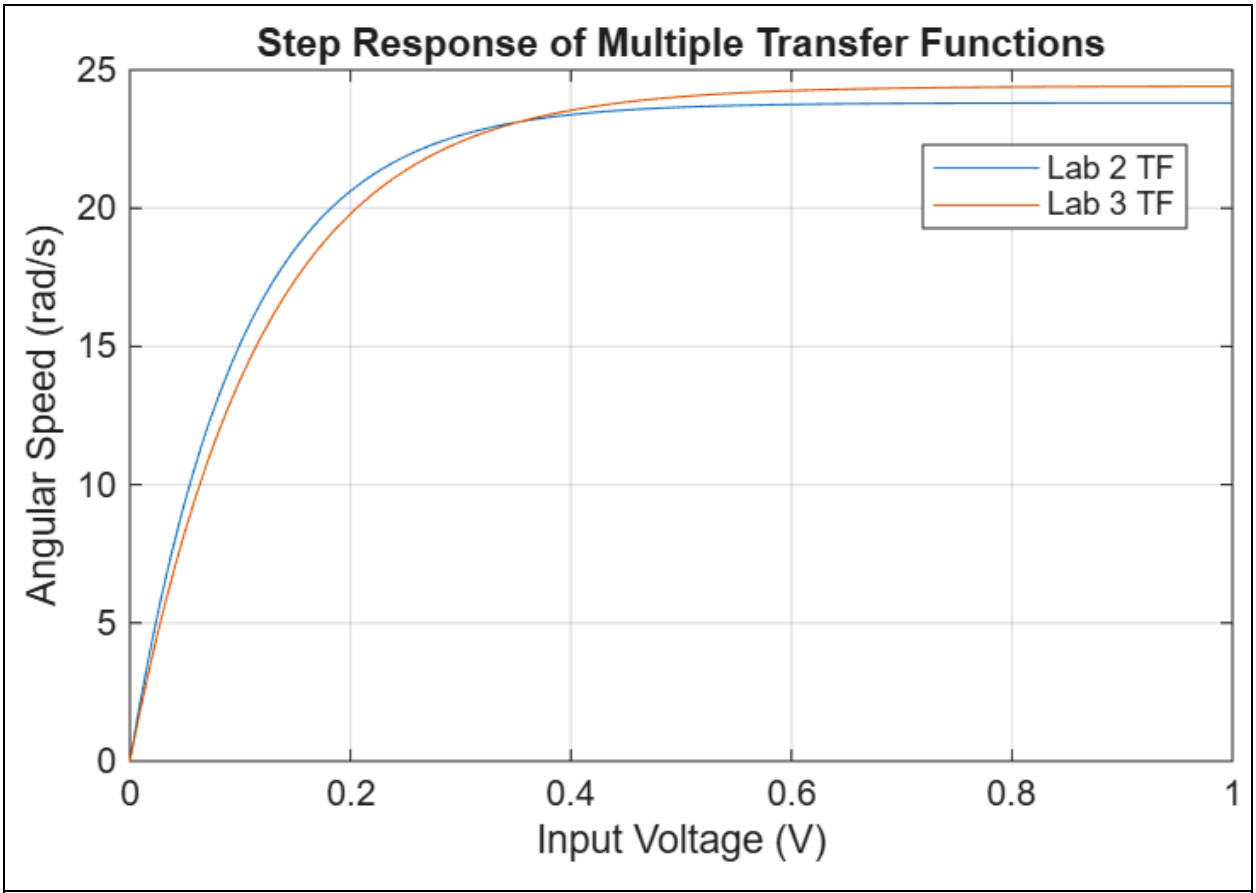


Figure 08: Plot of Step Response to Different Transfer Functions

The following is the derivation for the transfer function for Voltage to Speed:

As derived from Lab 2:

$$\frac{d\omega_m(t)}{dt} = \frac{k_t}{R_m \times J_{eq}} [v_m(t) - k_m \omega_m(t)]$$

$$\frac{d\omega_m(t)}{dt} = \frac{k_t}{R_m \times J_{eq}} v_m(t) - \frac{k_t \times k_m}{R_m \times J_{eq}} \omega_m(t)$$

Taking the Laplace Transform of this differential equation:

$$s \bullet \Omega_m(s) = \frac{k_t}{R_m \times J_{eq}} V_m(s) - \frac{k_t \times k_m}{R_m \times J_{eq}} \Omega_m(s)$$

We can then determine the transfer function, $\frac{\Omega_m(s)}{V_m(s)}$:

$$\Omega_m(s)\left[s + \frac{k_t \times k_m}{R_m \times J_{eq}}\right] = \frac{k_t}{R_m \times J_{eq}} V_m(s)$$

$$\Omega_m(s)\left[\frac{R_m \times J_{eq}}{k_t} s + k_m\right] = V_m(s)$$

$$T(s) = \frac{\Omega_m(s)}{V_m(s)} = \frac{1}{\left[\frac{R_m \times J_{eq}}{k_t} s + k_m\right]}$$

In Lab 2, we used the following values:

$$R_m = 8.4\Omega,\ k_t = 0.042\ N.m/A,\ \ k_m = 0.042\ V/(rad/s),\ J_{eq} = 2.089856 \times 10^{-5}\ kg.m^2$$

Substituting in our values into our transfer function:

$$T(s)_{(Lab\ 2)} = \frac{\Omega_m(s)}{V_m(s)} = \frac{1}{[0.004179712s + 0.042]}$$

In Lab 3, we obtain the following transfer function using our experimentally determined values for R_m and k_m:

$$R_m = 9.8862\Omega,\ k_t = 0.042\ N.m/A,\ \ k_m = 0.040961 V/(rad/s),\ J_{eq} = 2.089856 \times 10^{-5}\ kg.m^2$$

$$T(s)_{(Lab\ 3\ -\ Experimental\ Values)} = \frac{\Omega_m(s)}{V_m(s)} = \frac{1}{[0.0049192225s + 0.040961]}$$

10. Comparing our experiment with the lab 3 experiment:

In our experiment in Part 1, we are taking the same steps as the Lab 3 manual to determine R_m through stalling the motor and setting $\omega_m(t) = 0$ rad/s. However, our experiment involves fewer iterations (3) than the experiment conducted in the lab.

Table 7: Our Experiment for Determining R_m

Repetitions	Voltage Input (V)		Current Output (A)	
1	$v_m(t)$	4	$i_m(t)$	0.4178
2			$i_m(t)$	0.4162
3			$i_m(t)$	0.4158
Average: $x_{average} = \frac{x_1 + \ldots + x_n}{n}$	$v_m(t)$	4	$i_m(t)$	0.4166

$$R_{m(our\ experiment)} = \frac{v_m(t)}{i_m(t)} = \frac{(4\ V)}{(0.4166\ A)} = 9.602\ \Omega$$

The value for R_m obtained using our experiment, when compared to the experiment outlined in the Lab 3 Manual, were very close. Recall from 6.3.6, the value calculated for R_m was 9.8862 Ω. This was expected as the method to determine R_m across both experiments was the same and the only difference was the number of iterations performed. Our experiment yielded a value for R_m approximately 2.87 % lower than the R_m determined in 6.3.7.

To determine k_m, our group performed a second independent experiment. We leveraged the KVL equation, $v_m(t) - R_m i_m(t) - k_m \omega_m(t) = 0$ to determine the value of k_m. The experiment involves using two different input voltages and a system of equations to find out k_m. We perform 3 iterations at each input voltage and take the average of the input parameters to find k_m:

$$v_{m(1)}(t) - R_m i_{m(1)}(t) - k_m \omega_{m(1)}(t) = 0$$
$$v_{m(2)}(t) - R_m i_{m(2)}(t) - k_m \omega_{m(2)}(t) = 0$$

Table 8: Trial #1 for determining k_m ($v_{m(1)}(t)$ = 4 V)

Repetition #	Current Output (A)		Angular Speed Outputs (rad/s)	
1	$i_{m(1,1)}(t)$	0.004029	$\omega_{m(1,1)}(t)$	94.91
2	$i_{m(1,2)}(t)$	0.005238	$\omega_{m(1,2)}(t)$	99.63
3	$i_{m(1,3)}(t)$	0.003223	$\omega_{m(1,3)}(t)$	96.81
Averages: $x_{average} = \frac{x_1 + \ldots + x_n}{n}$	$i_{m(1)}(t)$	0.004163	$\omega_{m(1)}(t)$	97.117

Table 9: Trial #2 for determining k_m ($v_{m(1)}(t)$ = 3 V)

Repetition #	Current Output (A)		Angular Speed Outputs (rad/s)	
1	$i_{m(2,1)}(t)$	0.004029	$\omega_{m(2,1)}(t)$	72.43
2	$i_{m(2,2)}(t)$	0.003223	$\omega_{m(2,2)}(t)$	71.86
3	$i_{m(2,3)}(t)$	0.004029	$\omega_{m(2,3)}(t)$	71.86
Averages: $x_{average} = \frac{x_1 + \ldots + x_n}{n}$	$i_{m(2)}(t)$	0.003760	$\omega_{m(2)}(t)$	72.05

$$k_{m(our\ experiment)} = \frac{\left[\frac{v_{m(2)}(t)}{i_{m(2)}(t)} - \frac{v_{m(1)}(t)}{i_{m(1)}(t)}\right]}{\left[\frac{\omega_{m(2)}(t)}{i_{m(2)}(t)} - \frac{\omega_{m(1)}(t)}{i_{m(1)}(t)}\right]} = 0.03912\ V/(rad/s)$$

The value for k_m obtained using our experiment, when compared to the experiment outlined in the Lab 3 Manual, were very close. Recall from 6.3.8, the value calculated for k_m was 0.040961 Ω. This was expected as even though the experiments were different, they used the same KVL equation and parameters (input voltage, output current and speed) to determine k_m. Our experiment yielded a value for k_m approximately 4.49 % lower than the k_m determined in 6.3.8.

Self Grade:

If we were to grade ourselves', we would give ourselves' a 10/10 because our lab procedure was sufficient in accurately yielding the parameters R_m and k_m separately.